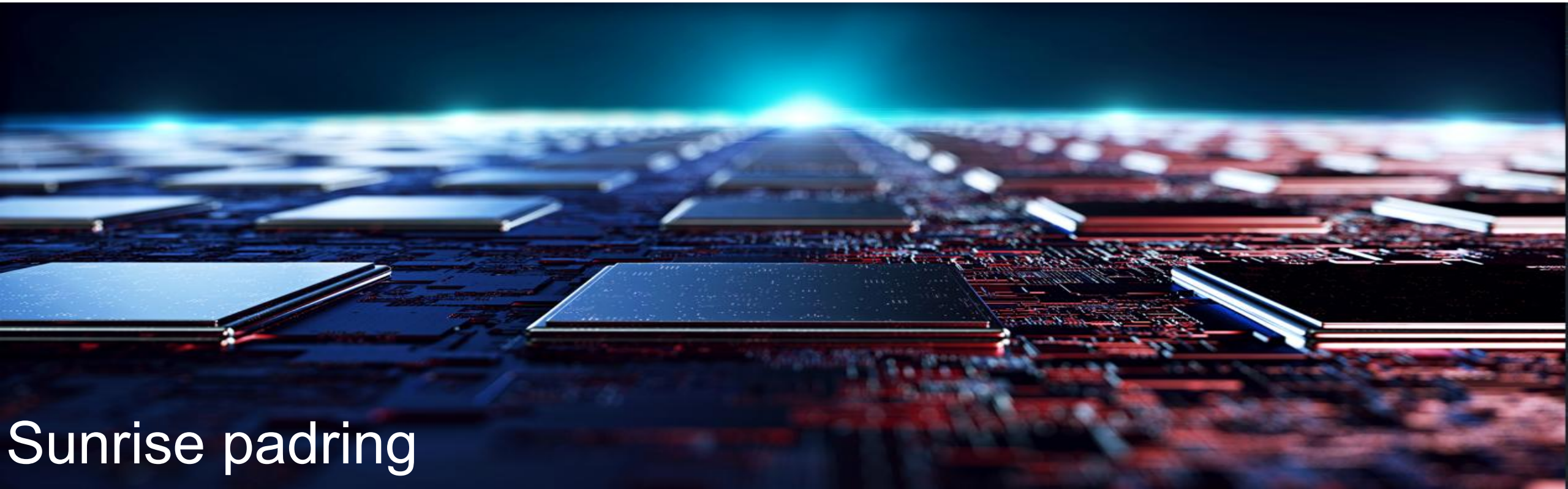




Sunrise padding

Diarmuid Collins

24.11.2025

Contents

- Summary / conclusions
- List of cells used
- Pinlist
- Analog IO
- Digital IO
- Digital tristate O
- Vddio pwr clamp
- Vssio cell
- Vdd pwr clamp
- Vss cell
- Complete padring
- Chip delivery based on padring

Summary / conclusions

- Full pin list has been defined
- I/O cells have been defined
- ESD has been verified for each defined I/O cell
- Each I/O cell has been laid out with a square bond pad of dimensions 80x80um
- Only I/O cell with DRCs is the analogIO (4 DRCs)
- However, IHP have confirmed these DRCs will be waived
- Full padding has been laid out
- Padding is DRC clean (apart from analog IO waivable DRCs)
- Total die area ~ 1mm²
- Excluding the padding leaves ~ 0.5x0.5mm of die area to fit the design into
- Initial estimates suggest this will be enough
- Individual I/O cell gds files, schematics and extracted netlists have been checked into github along with the completed padding
- github location is: <https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padding>

List of cells used

- All io cells can be found at: [\\$PDK_ROOT/ihp-sg13g2/libs.ref/sg13g2_io/gds](#)

Cells
▼ sg12g2_Gallery
sg13g2_Corner
sg13g2_Filler1000
sg13g2_Filler10000
sg13g2_Filler200
sg13g2_Filler2000
sg13g2_Filler400
sg13g2_Filler4000
▶ sg13g2_IOPadAnalog
▶ sg13g2_IOPadIOVdd
▶ sg13g2_IOPadIOVss
▶ sg13g2_IOPadIn
▶ sg13g2_IOPadInOut16mA
▶ sg13g2_IOPadInOut30mA
▶ sg13g2_IOPadInOut4mA
▶ sg13g2_IOPadOut16mA
▶ sg13g2_IOPadOut30mA
▶ sg13g2_IOPadOut4mA
▶ sg13g2_IOPadTriOut16mA
▶ sg13g2_IOPadTriOut30mA
▶ sg13g2_IOPadTriOut4mA
▶ sg13g2_IOPadVdd
▶ sg13g2_IOPadVss

- Highlighted cells (left) are used in the Sunrise padring
- Note the “30mA” on the digital i/o’s represents the cells short circuit current sustainability
- All iocells + 80x80um bondpad used have been checked into the Sunrise repo at:

<https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring>

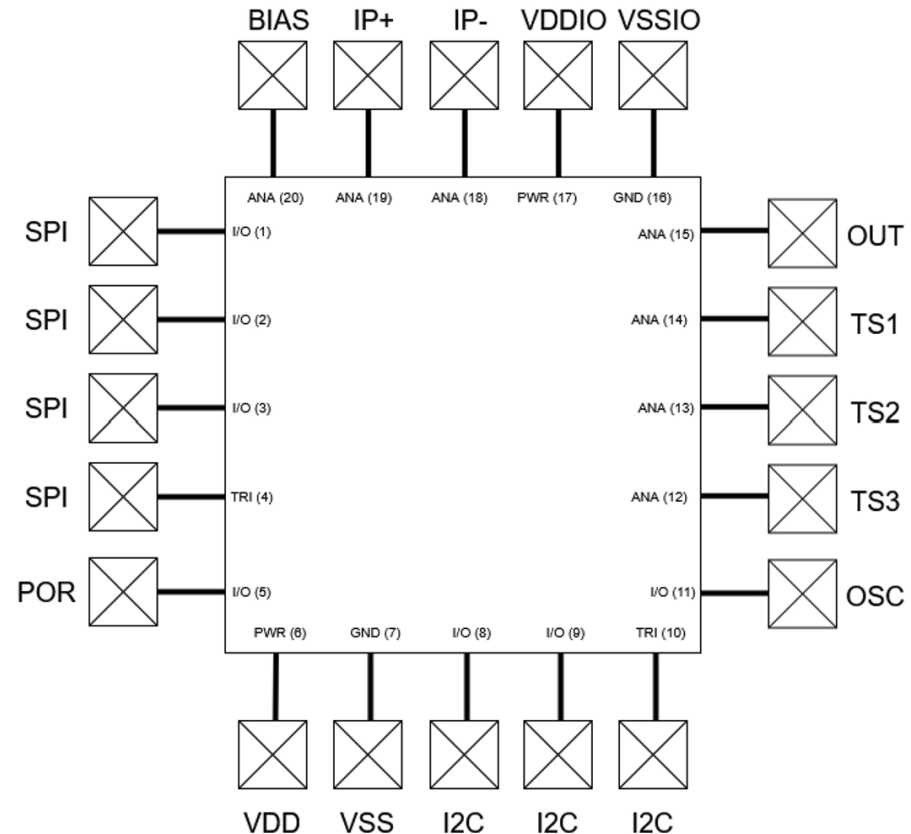
- Below shows the layout view of all listed cells



Pinlist

- Below shows the pinlist and their positions on the package

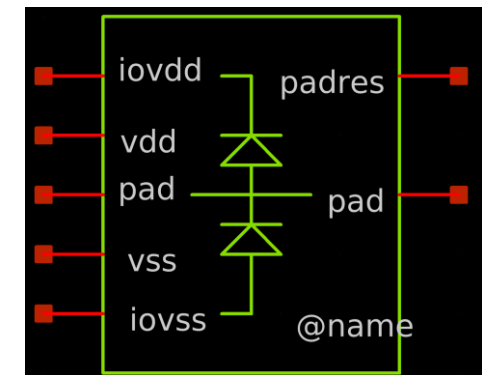
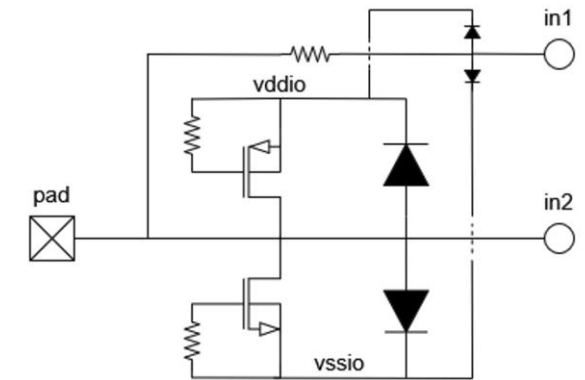
Pin #	NAME	TYPE	DESCRIPTION
1	SPI	DIGIO	SPI ctrl
2	SPI	DIGIO	SPI ctrl
3	SPI	DIGIO	SPI ctrl
4	SPI	DIGTRI	SPI ctrl
5	POR	DIGIO	Power on reset
6	VDD	PWR	Digital supply (1v5)
7	VSS	PWR	Digital ground
8	I2C	DIGIO	I2C ctrl
9	I2C	DIGIO	I2C ctrl
10	I2C	DIGTRI	I2C ctrl
11	OSC	DIGIO	Oscillation output
12	TS3	ANAIO	Temp sense output
13	TS2	ANAIO	Temp sense output
14	TS1	ANAIO	Temp sense output
15	OUT	ANAIO	INA output
16	VSSIO	PWR	Analog ground
17	VDDIO	PWR	Analog supply (3v3)
18	IP-	ANAIO	INA input
19	IP+	ANAIO	INA input
20	BIAS	ANAIO	INA bias



- A QFN-20 package has currently been ordered at IHP to accommodate this

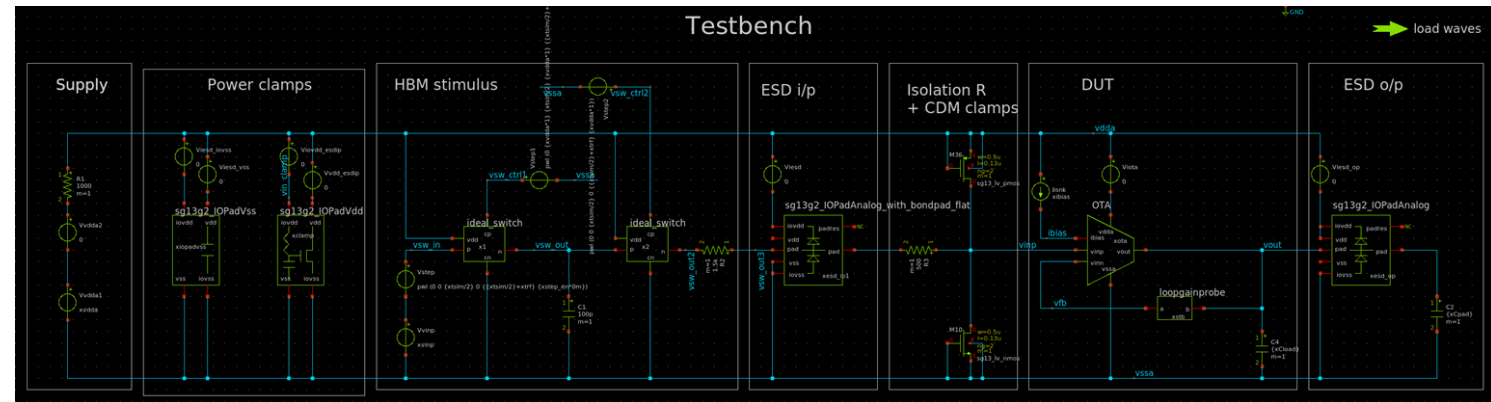
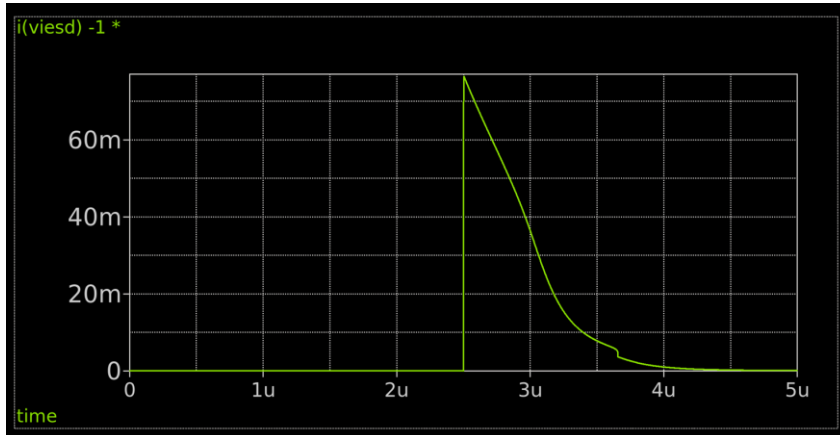
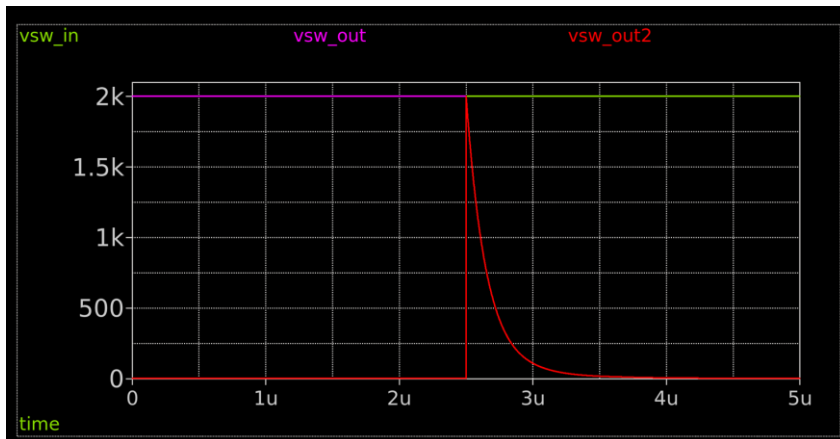
AnalogIO: Schematic

- AnalogIO (with 80x80um bondpad) has been checked into:
- <https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/IOPadAnalog>
- Right shows the schematic drawing of the cell where it can be seen to have two outputs
- “in1” has an isolation resistor and b2b diodes
- “in2” has b2b snapback devices in parallel with b2b diodes
- We will use “in2” for the added protection of the snapbacks
- Isolation resistor (0.5 / 1k) will be placed separately in layout followed by CDM clamps placed locally at the inputs to the relevant analog block
- Right shows the schematic symbol which can be found at the above link



AnalogIO: ESD verification

- To verify ESD, the below HBM setup was created where 2kV was discharged into the I/O through a 1.5k resistor in series with an 100pF cap

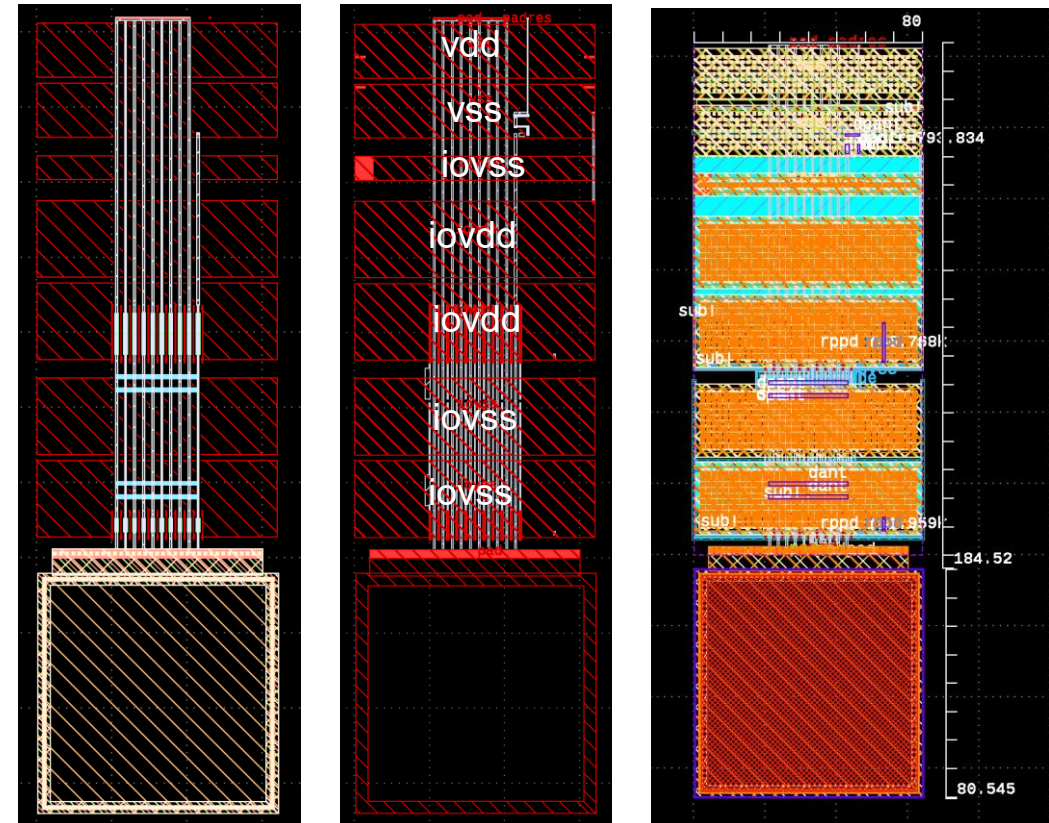


- From 0 → 2.5us, 100pF cap is charged to 2kV
- At 2.5us, the cap is discharged into the I/O through the 1.5k resistor
- Switch output (vsw_out2) can be clearly seen to discharge
- Current from the I/O to the power rail is shown as i(viesd)
- This clearly shows the discharge path from input to the power rail to activate during an ESD event

AnalogIO: Layout

- Right shows the layout of the I/O cell including the bondpad
- I/O cell footprint = 80 x 180um
- Bondpad footprint = 80 x 80um
- I/O cell + bondpad footprint = 80 x 260um
- Output pin occurs on M3
- M3 view clearly shows the supply connections
- M3 view shows the connectivity from pad to o/p
- Highlighting the bondpad clearly shows the “in2” path through the cell
- gds can be found in the git repo as:

 [sg13g2_IOPadAnalog_with_bondpad.gds](#)



AnalogIO: DRC

- 4 DRCs remain in the block as shown

Database	sg13g2_IOPadAnalog_with_bondpad_sg13g2_IOPadAnalog_full	
... on layout	sg13g2_IOPadAnalog_with_bondpad.gds	
▶		
Directory		
Cell / Category	▲	Count (Not Visited) - Waived
▼ By Cell		4 (4)
[sg13g2_IOPadAnalog]		
▼ [sg13g2_Clamp_P20N0D]		4 (4)
Sdiod.d		2 (2)
Sdiod.e		2 (2)
▼ By Category		4 (4)

- These however can be waived according to:
<https://github.com/IHP-GmbH/IHP-Open-PDK/issues/684>
- Therefore, effectively, the block is DRC clean

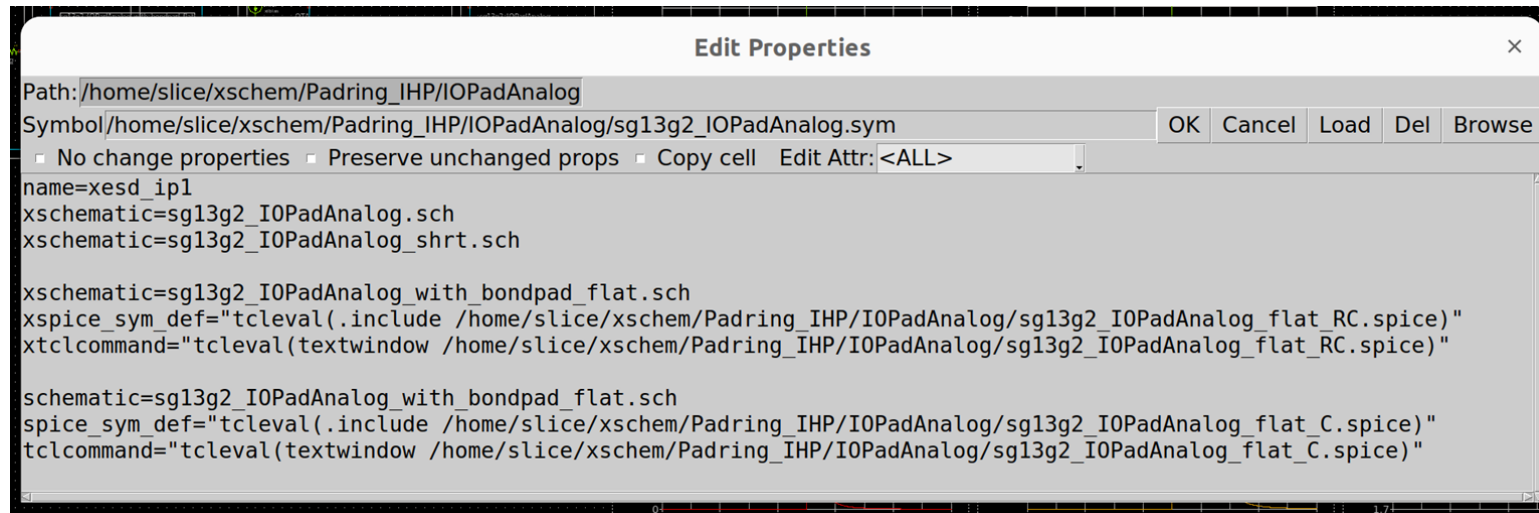
AnalogIO: PEX

- PEX C / RC files have been checked into the git repo as:

 sg13g2_IOPadAnalog_flat_C.spice

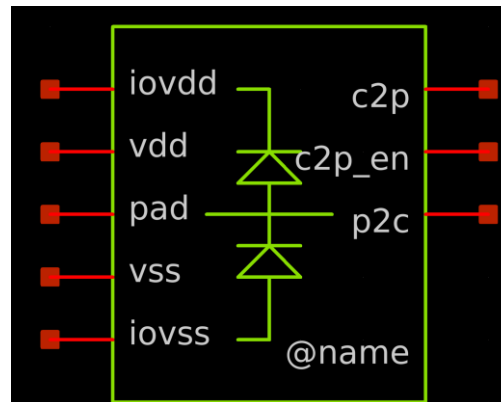
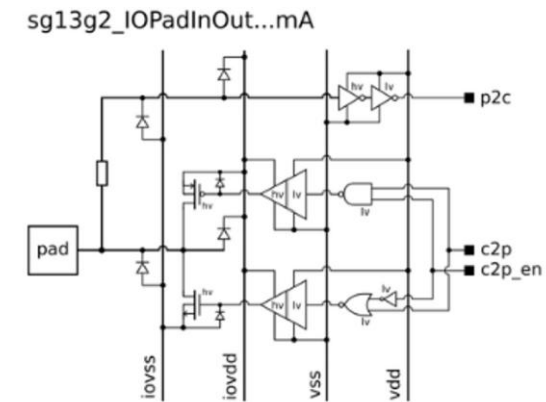
 sg13g2_IOPadAnalog_flat_RC.spice

- These can be cloned into a local area and run in xschem with instance binding similar to the below:



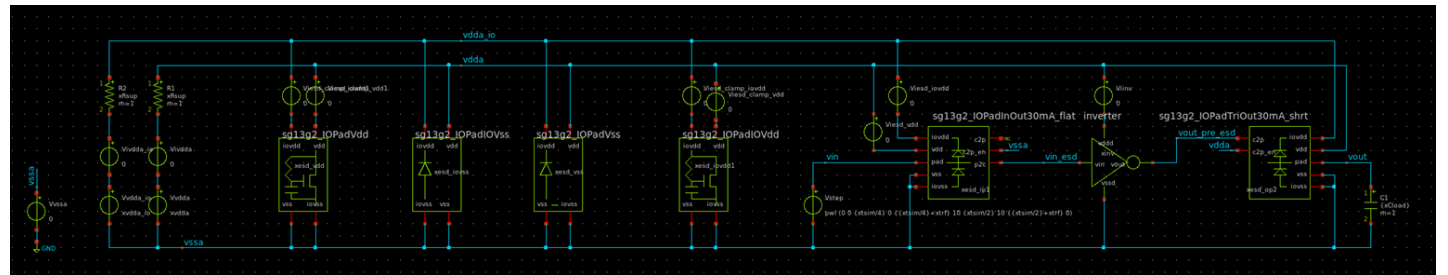
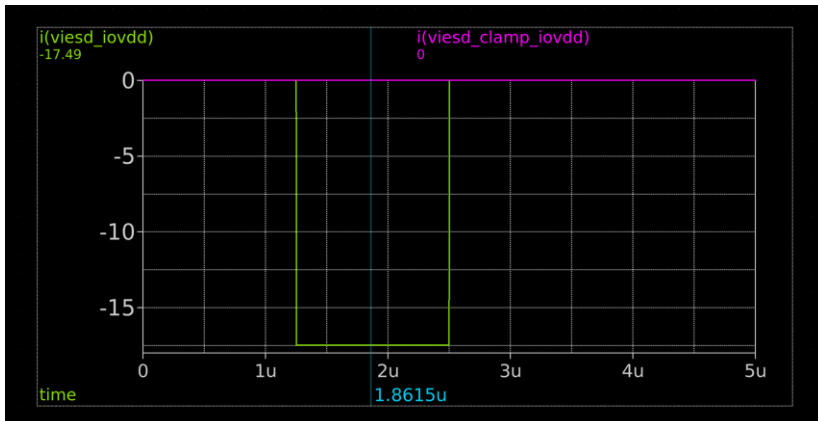
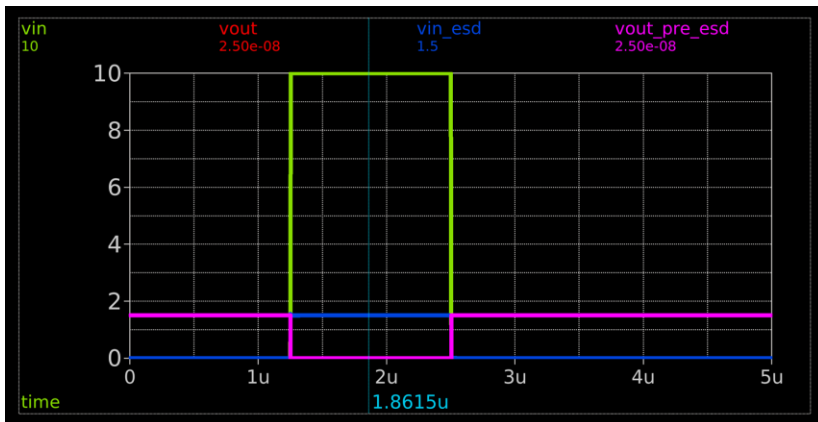
DigitalIO: Schematic

- DigitalIO (with 80x80um bondpad) has been checked into:
- https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/IOPadDigital_30mA
- Right shows the schematic drawing of the cell where it can be seen to be configurable as i/p (pad to chip – p2c) and an o/p (chip to pad – c2p)
- To configure the cell as i/p, c2p_en is set to vss (1v5)
- To configure the cell as an o/p, c2p_en is set to vdd
- Both i/p and o/p configurations use b2b diodes for protection
- Below shows the schematic symbol which can be found at the above link



DigitalIO: ESD verification

- To verify ESD, a 10V step was applied to the i/p of the cell and the current through it monitored

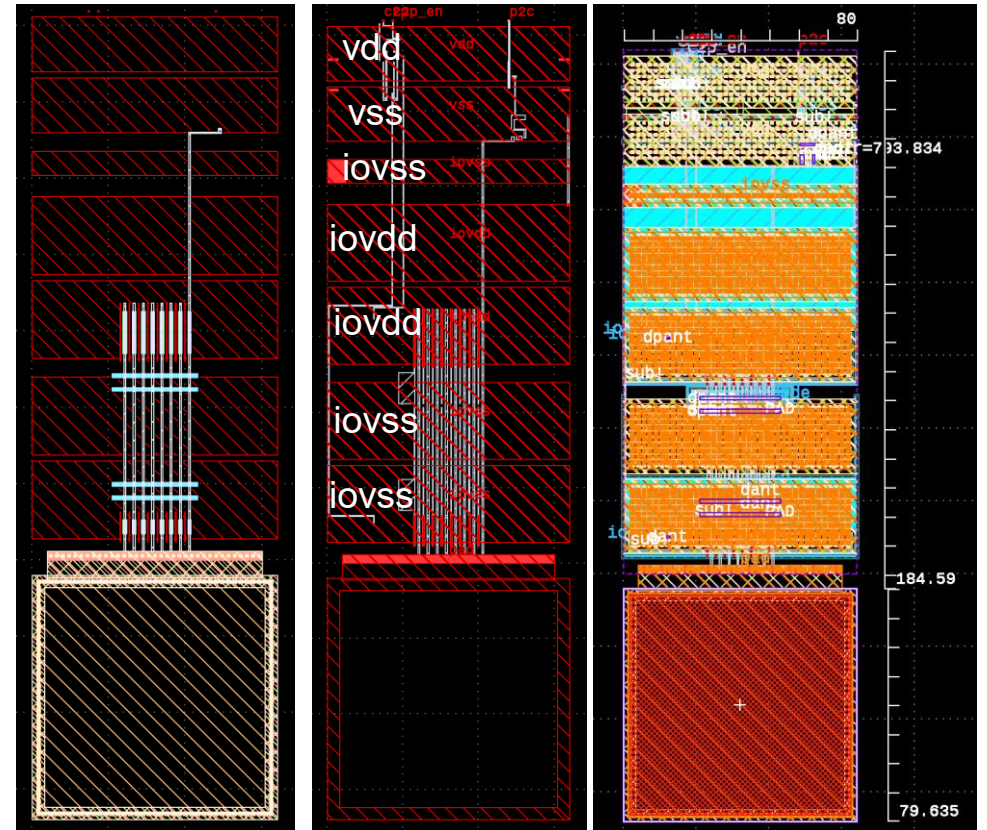


- At 1.25us a 10V step input is applied
- Current from the I/O to the power rail is shown as $i(viesd_iovd)$
- This clearly shows the discharge path from input to the power rail to activate during an ESD event

DigitalIO: Layout

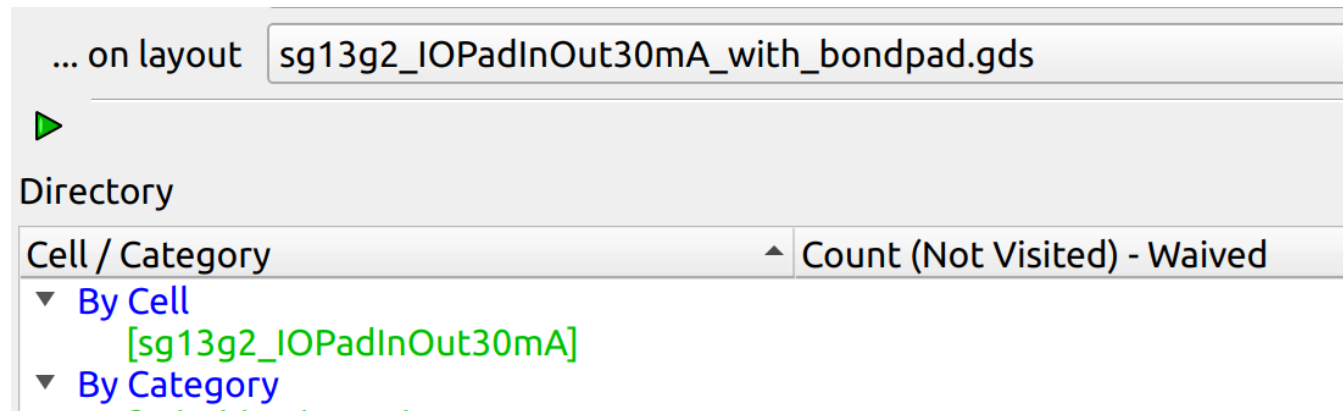
- Right shows the layout of the I/O cell including the bondpad
- I/O cell footprint = 80 x 180um
- Bondpad footprint = 80 x 80um
- I/O cell + bondpad footprint = 80 x 260um
- Output pin occurs on M3
- M3 view clearly shows the supply connections
- M3 view clearly shows the connectivity from p2c and c2p through through the cell
- gds can be found in the git repo as:

 sg13g2_IOPadInOut30mA_with_bondpad.gds



DigitalIO: DRC

- Block is DRC clean

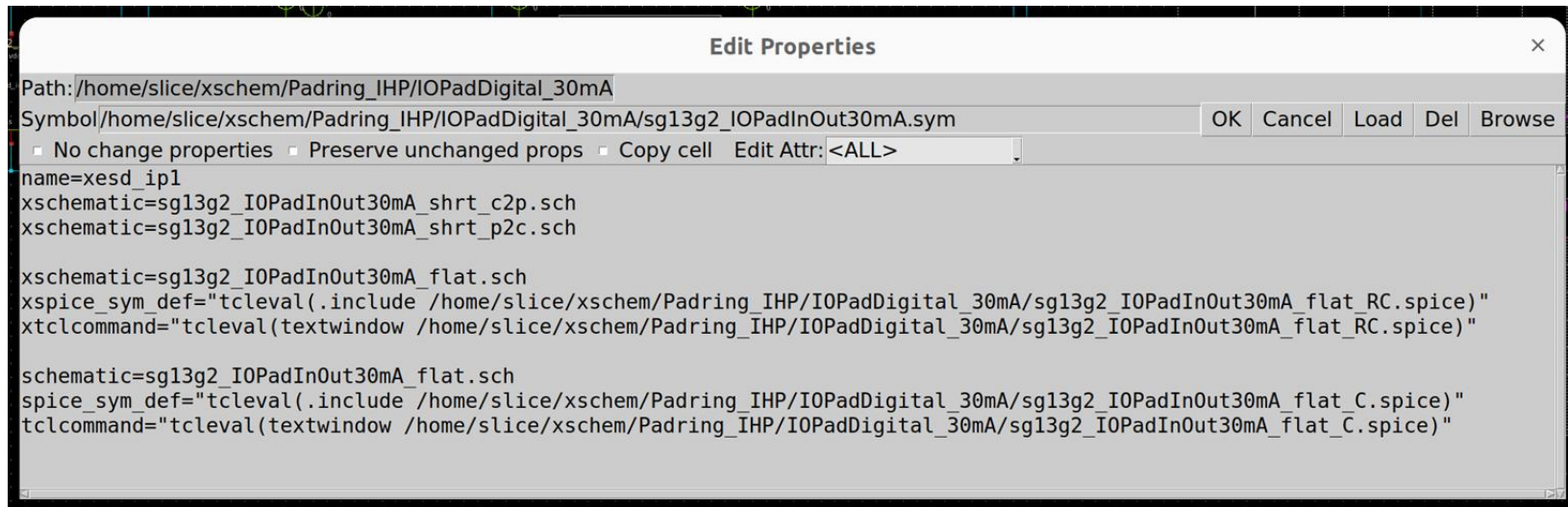


DigitalIO: PEX

- PEX C / RC files have been checked into the git repo as:

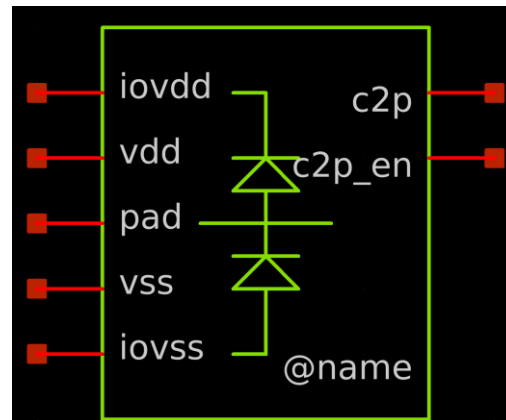
```
sg13g2_IOPadInOut30mA_flat_C.spice
sg13g2_IOPadInOut30mA_flat_RC.spice
```

- These can be cloned into a local area and run in xschem with instance binding similar to the below:

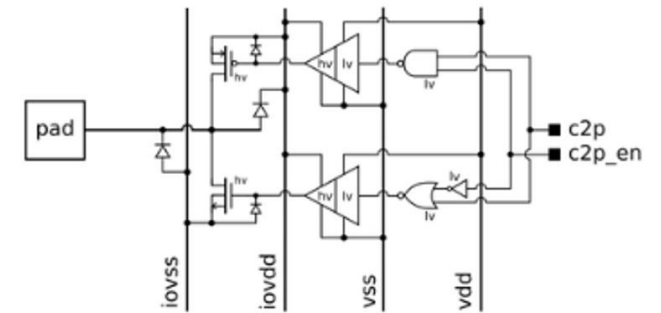


DigitalTristateO: Schematic

- DigitalTristateO (with 80x80um bondpad) has been checked into:
- https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/IOTriPadDigital_30mA
- Right shows the schematic drawing of the cell where it can be seen to be configurable as an o/p (chip to pad – c2p) and as a tri-state output
- To configure the cell as an o/p, c2p_en is set to vdd
- To put the cell in hiZ, c2p_en is set to vss
- O/p configuration uses b2b diodes for protection
- Below shows the schematic symbol which can be found at the above link



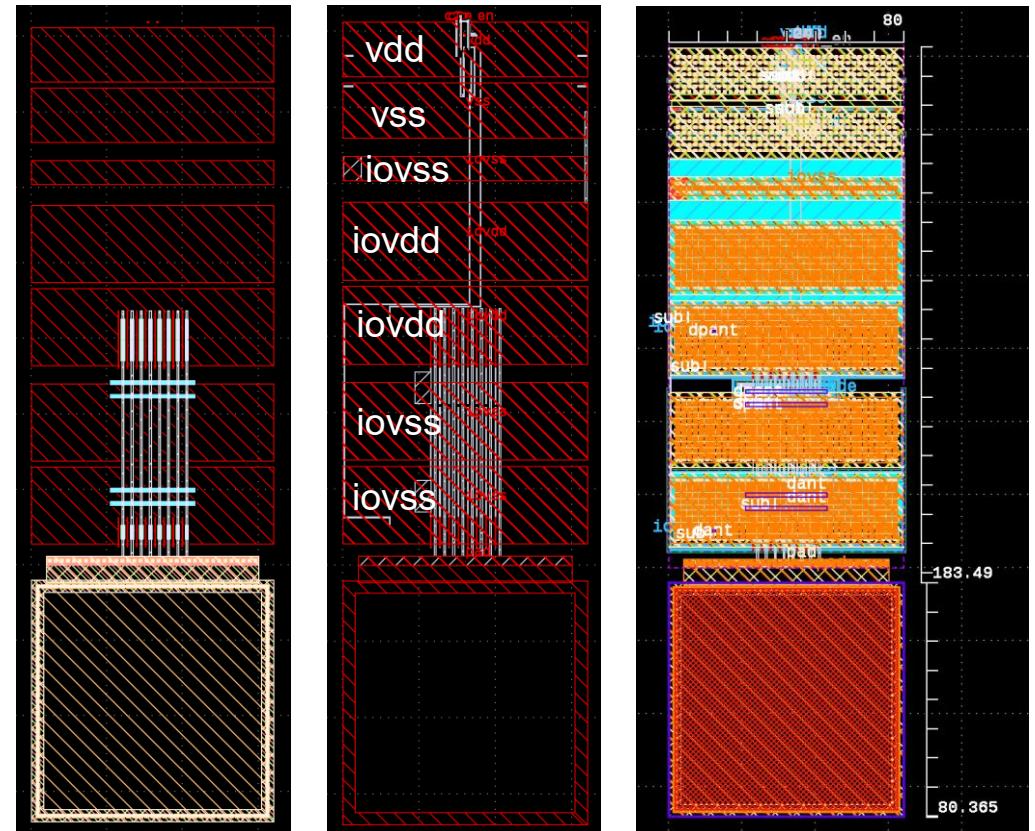
sg13g2_IOPadTriOut...mA



DigitalTristateO: Layout

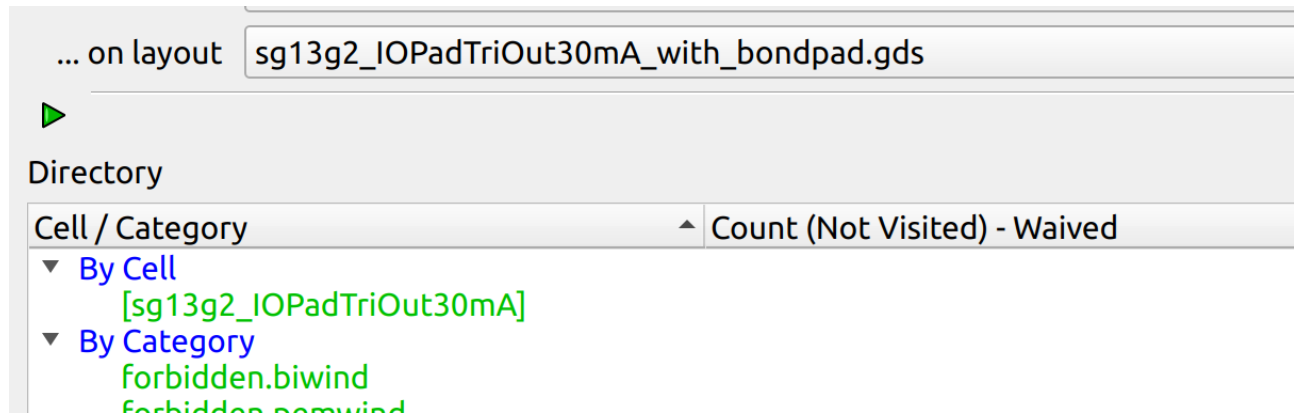
- Right shows the layout of the I/O cell including the bondpad
- I/O cell footprint = 80 x 180um
- Bondpad footprint = 80 x 80um
- I/O cell + bondpad footprint = 80 x 260um
- Output pin occurs on M3
- M3 view clearly shows the supply connections
- M3 view clearly shows the connectivity from c2p through through the cell
- gds can be found in the git repo as:

 sg13g2_IOPadTriOut30mA_with_bondpad.gds



DigitalTristateO: DRC

- Block is DRC clean



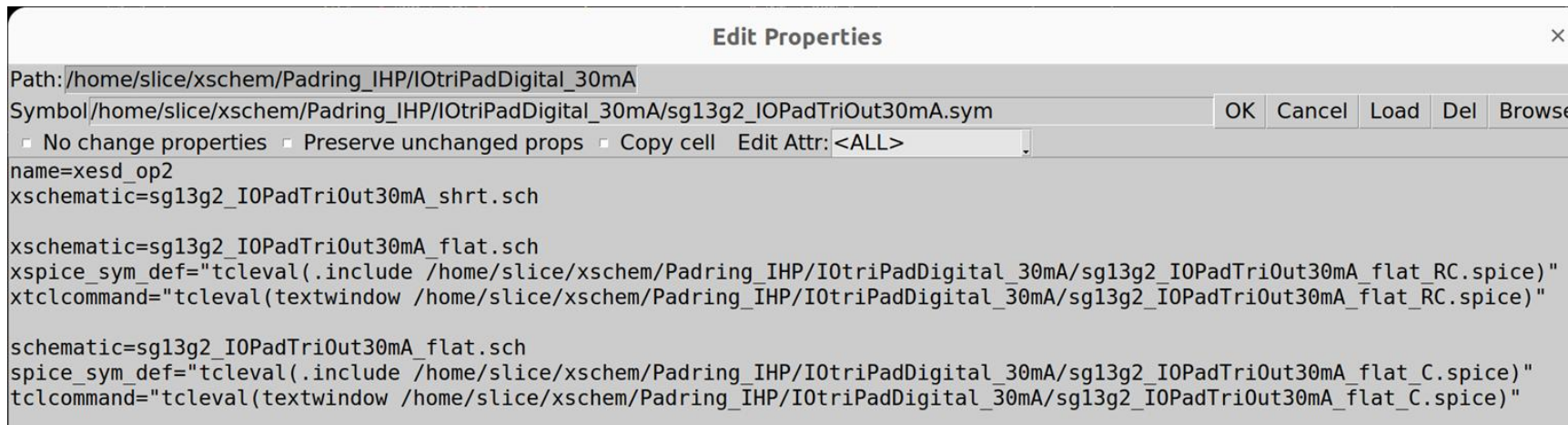
DigitalTristateO: PEX

- PEX C / RC files have been checked into the git repo as:

 sg13g2_IOPadTriOut30mA_flat_C.spice

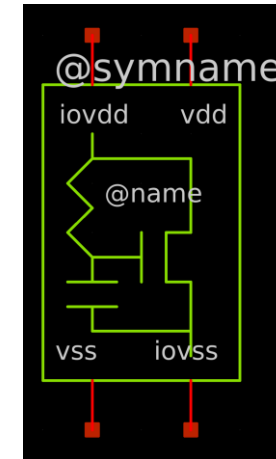
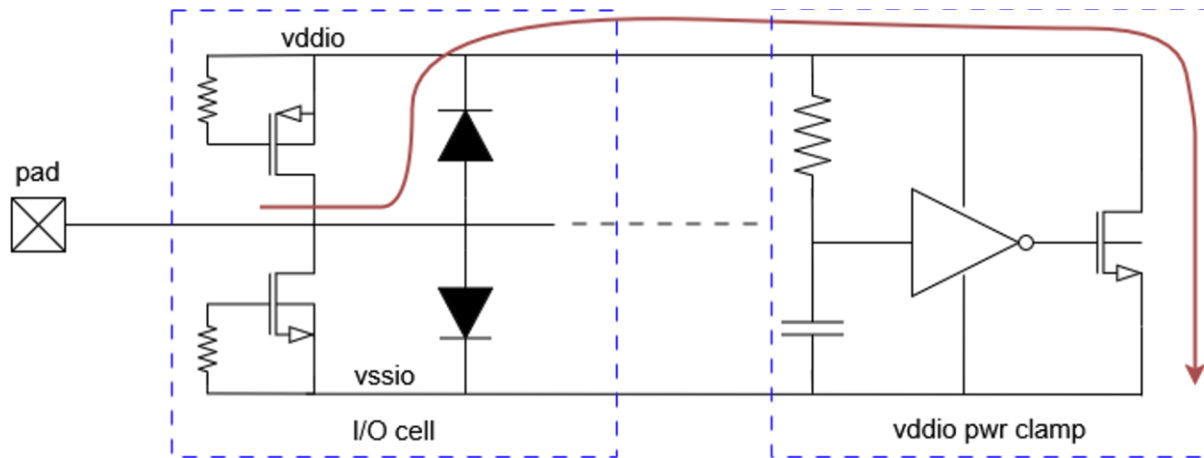
 sg13g2_IOPadTriOut30mA_flat_RC.spice

- These can be cloned into a local area and run in xschem with instance binding similar to the below:



Vddio pwr clamp: Schematic

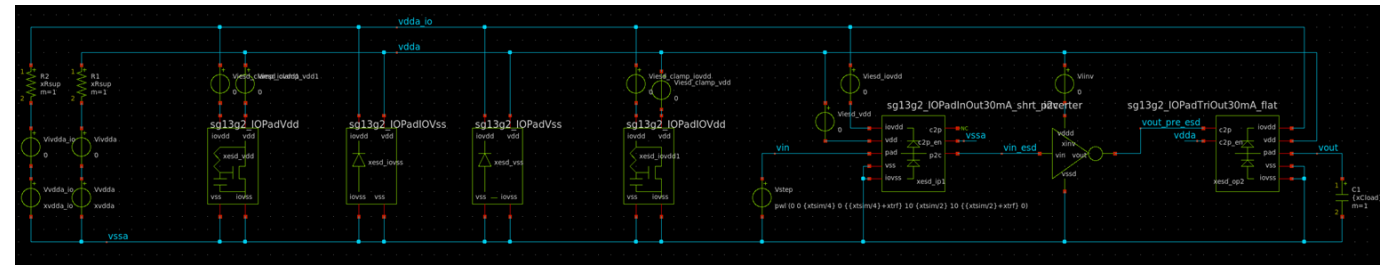
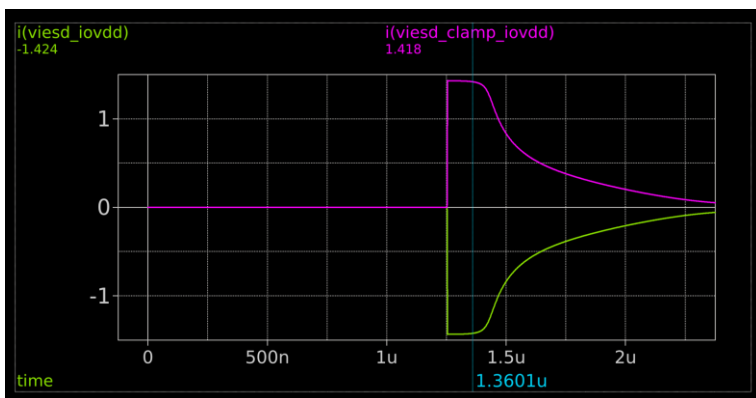
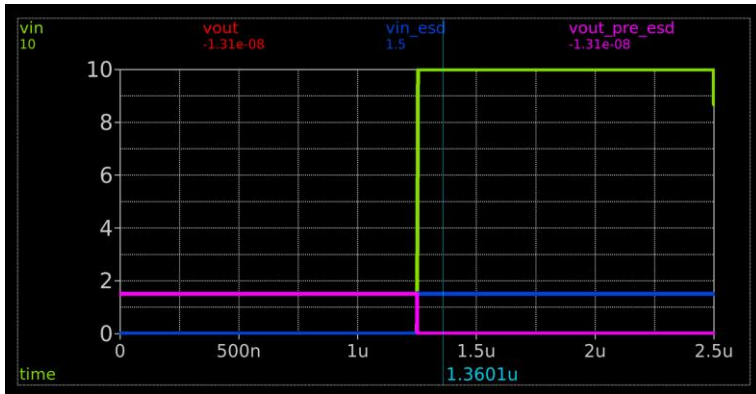
- Vddio pwr clamp (with 80x80um bondpad) has been checked into:
- https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/vddio_pwr_clamp
- Schematic / symbol of the clamp is shown below



- As shown, charge from an ESD event gets shunted onto vddio through the I/O cell
- Voltage builds up across the impedance of vddio. Due to the RC time constant seen at the inverters input, a vgs develops across the inverters PMOS to momentarily turn it on. This turns on the shunting NMOS (or clamp) to discharge the ESD event from vddio to vssio. This also discharges the inverters PMOS' vgs. As a result the NMOS clamp turns off once the event has been discharged.

Vddio pwr clamp: ESD verification

- To verify ESD, a 10V step was applied to the i/p of an I/O cell and the current through the clamp monitored

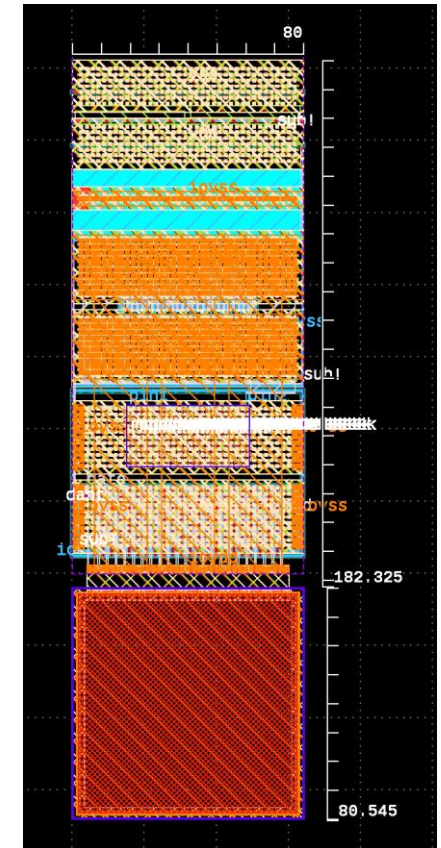
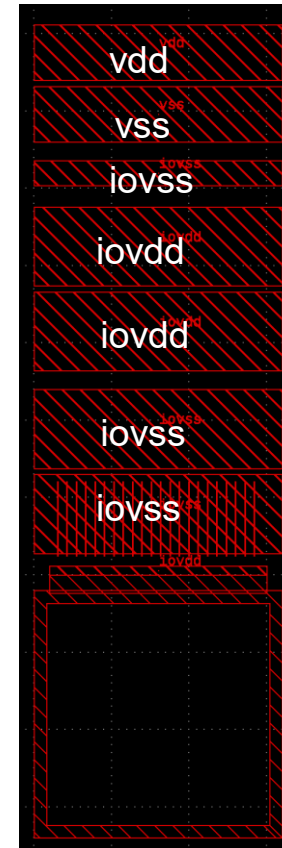


- At 1.25us a 10V step input is applied
- Current from the I/O to vddio is shown as i(viesd_iovdd)
- Current from the vddio through the clamp is shown as i(viesd_clamp_iovdd)
- This clearly shows the discharge path from input to the vddio through the I/O cell, and from vddio to vssio through the clamp
- Note: An impedance was placed on the power supply to force charge through the clamp in simulation

Vddio pwr clamp: Layout

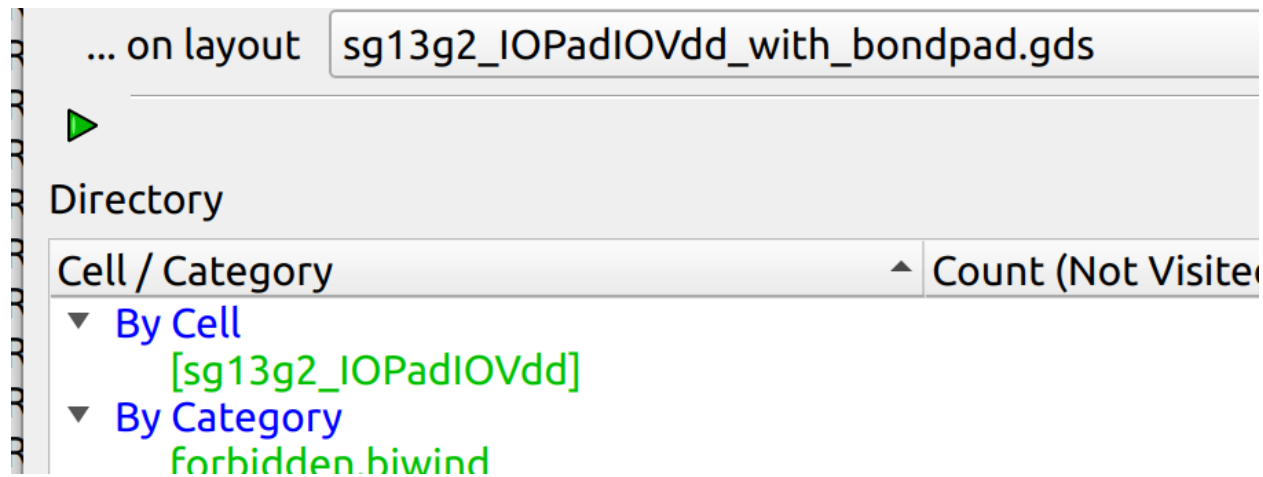
- Right shows the layout of the vddio pwr cell including the bondpad
- I/O cell footprint = 80 x 180um
- Bondpad footprint = 80 x 80um
- I/O cell + bondpad footprint = 80 x 260um
- Output pin occurs on M3
- M3 view clearly shows the supply connections
- gds can be found in the git repo as:

 sg13g2_IOPadIOVdd_with_bondpad.gds



Vddio pwr clamp: DRC

- Block is DRC clean



Vddio pwr clamp: PEX

- PEX C / RC files have been checked into the git repo as:

 sg13g2_IOPadIOVdd_flat_C.spice

 sg13g2_IOPadIOVdd_flat_RC.spice

- These can be cloned into a local area and run in xschem with instance binding similar to the below:

```
Path:/home/slice/xschem/Padring_IHP/vddio_pwrclamp
Symbol/home/slice/xschem/Padring_IHP/vddio_pwrclamp/sg13g2_IOPadIOVdd.sym OK Cancel Load
  No change properties  Preserve unchanged props  Copy cell  Edit Attr: <ALL>
name=xesd_iovdd1

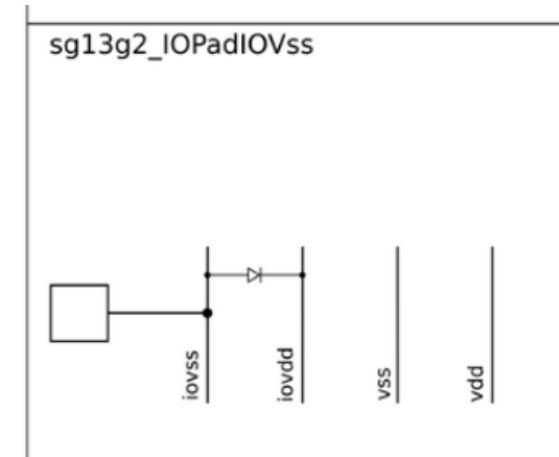
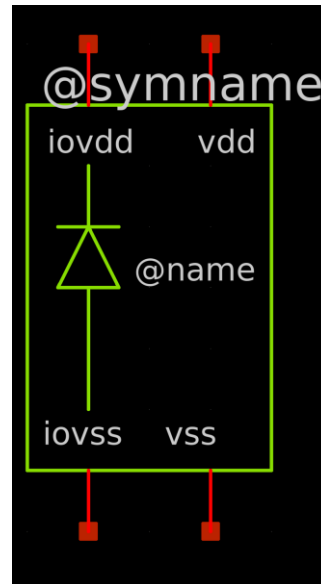
xschematic=sg13g2_IOPadIOVdd.sch

xschematic=sg13g2_IOPadIOVdd_flat.sch
xspice_sym_def="tcleva(.include /home/slice/xschem/Padring_IHP/vddio_pwrclamp/sg13g2_IOPadIOVdd_flat_RC.spice)"
xtclcommand="tcleva(textwindow /home/slice/xschem/Padring_IHP/vddio_pwrclamp/sg13g2_IOPadIOVdd_flat_RC.spice)"

schematic=sg13g2_IOPadIOVdd_flat.sch
spice_sym_def="tcleva(.include /home/slice/xschem/Padring_IHP/vddio_pwrclamp/sg13g2_IOPadIOVdd_flat_C.spice)"
tclcommand="tcleva(textwindow /home/slice/xschem/Padring_IHP/vddio_pwrclamp/sg13g2_IOPadIOVdd_flat_C.spice)"
```

Vssio cell: Schematic

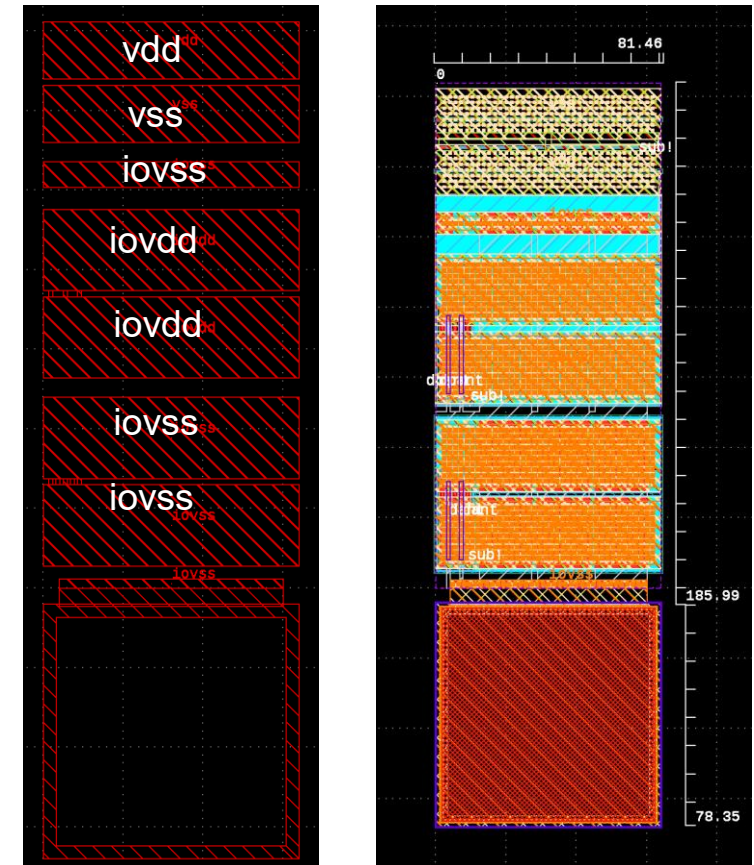
- Vssio cell (with 80x80um bondpad) has been checked into:
- https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/vssio_clamp
- Right shows the schematic drawing of the cell where it can be seen to be a basic cell consisting of a diode between iovss and iovdd
- elow shows the schematic symbol which can be found at the above link



Vssio pwr clamp: Layout

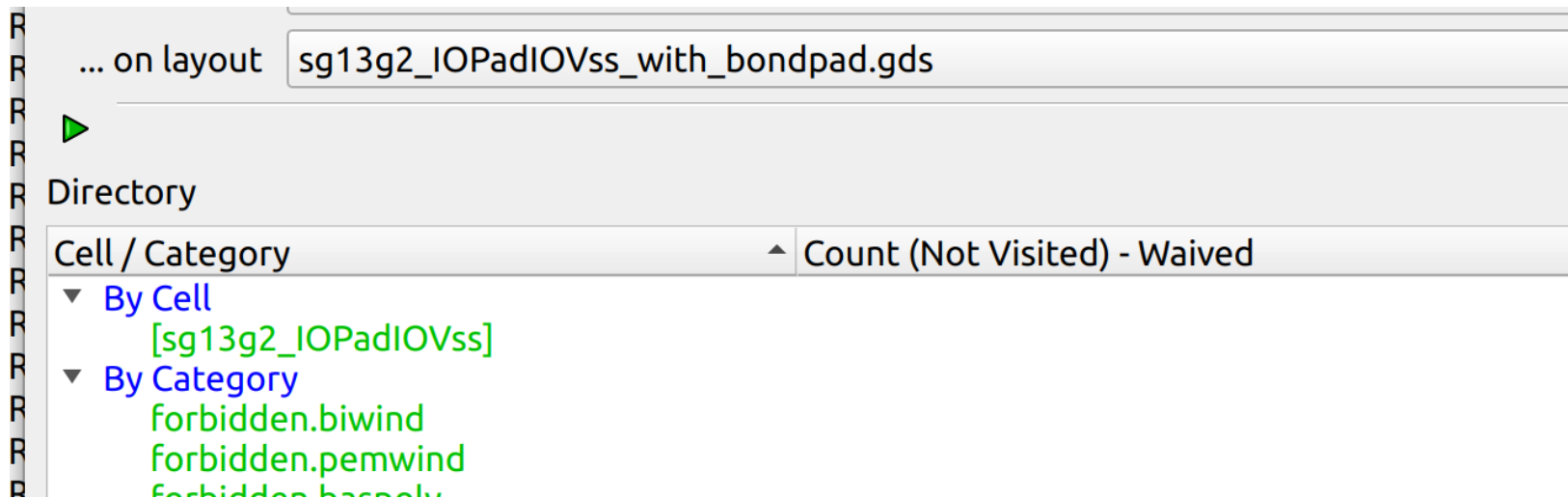
- Right shows the layout of the vddio pwr cell including the bondpad
- I/O cell footprint = 80 x 180um
- Bondpad footprint = 80 x 80um
- I/O cell + bondpad footprint = 80 x 260um
- Output pin occurs on M3
- M3 view clearly shows the supply connections
- gds can be found in the git repo as:

 sg13g2_IOPadIOVss_with_bondpad.gds



Vssio pwr clamp: DRC

- Block is DRC clean



Vssio pwr clamp: PEX

- PEX C / RC files have been checked into the git repo as:

 sg13g2_IOPadIOVss_flat_C.spice

 sg13g2_IOPadIOVss_flat_RC.spice

- These can be cloned into a local area and run in xschem with instance binding similar to the below:

```
Path: /home/slice/xschem/Padring_IHP/vssio_clamp
Symbol /home/slice/xschem/Padring_IHP/vssio_clamp/sg13g2_IOPadIOVss.sym OK Cancel Load Del Br
  ■ No change properties ■ Preserve unchanged props ■ Copy cell Edit Attr: <ALL>
name=xesd_iovss

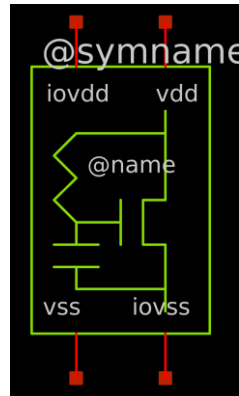
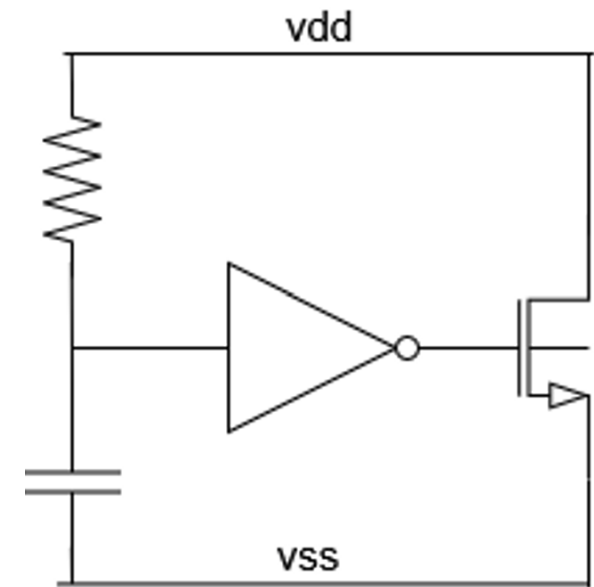
schematic=sg13g2_IOPadIOVss.sch

xschematic=sg13g2_IOPadIOVss_flat.sch
xspice_sym_def="tcleva(.include /home/slice/xschem/Padring_IHP/vssio_clamp/sg13g2_IOPadIOVss_flat_RC.spice)"
xtclcommand="tcleva(textwindow /home/slice/xschem/Padring_IHP/vssio_clamp/sg13g2_IOPadIOVss_flat_RC.spice)"

xschematic=sg13g2_IOPadIOVss_flat.sch
xspice_sym_def="tcleva(.include /home/slice/xschem/Padring_IHP/vssio_clamp/sg13g2_IOPadIOVss_flat_C.spice)"
xtclcommand="tcleva(textwindow /home/slice/xschem/Padring_IHP/vssio_clamp/sg13g2_IOPadIOVss_flat_C.spice)"
```

Vdd pwr clamp: Schematic

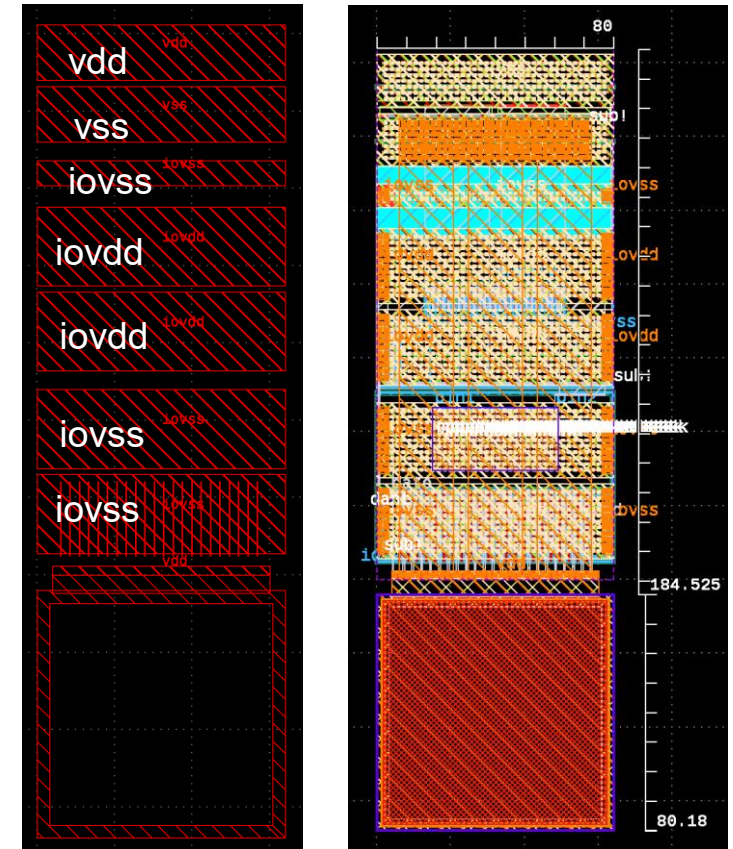
- Vdd pwr clamp (with 80x80um bondpad) has been checked into:
- https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/vdd_pwr_clamp
- Right shows the schematic drawing of the cell where it can be seen to be the standard clamp configuration seen for vddio
- Any ESD charge routed onto vdd thus gets shunted to vss through the large FET in the clamp
- How useful is this clamp is debatable since most, if not all, the charge from an ESD event gets shunted onto the vddio rail
- Below shows the schematic symbol which can be found at the above link



Vdd pwr clamp: Layout

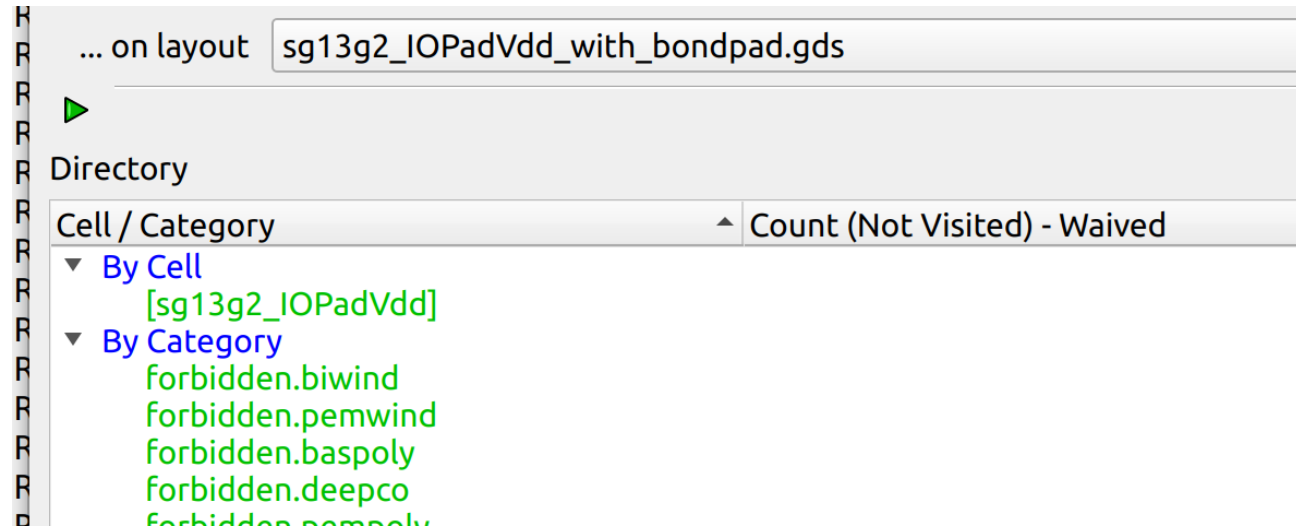
- Right shows the layout of the vddio pwr cell including the bondpad
- I/O cell footprint = 80 x 180um
- Bondpad footprint = 80 x 80um
- I/O cell + bondpad footprint = 80 x 260um
- Output pin occurs on M3
- M3 view clearly shows the supply connections
- gds can be found in the git repo as:

 sg13g2_IOPadVdd_with_bondpad.gds



Vdd pwr clamp: DRC

- Block is DRC clean



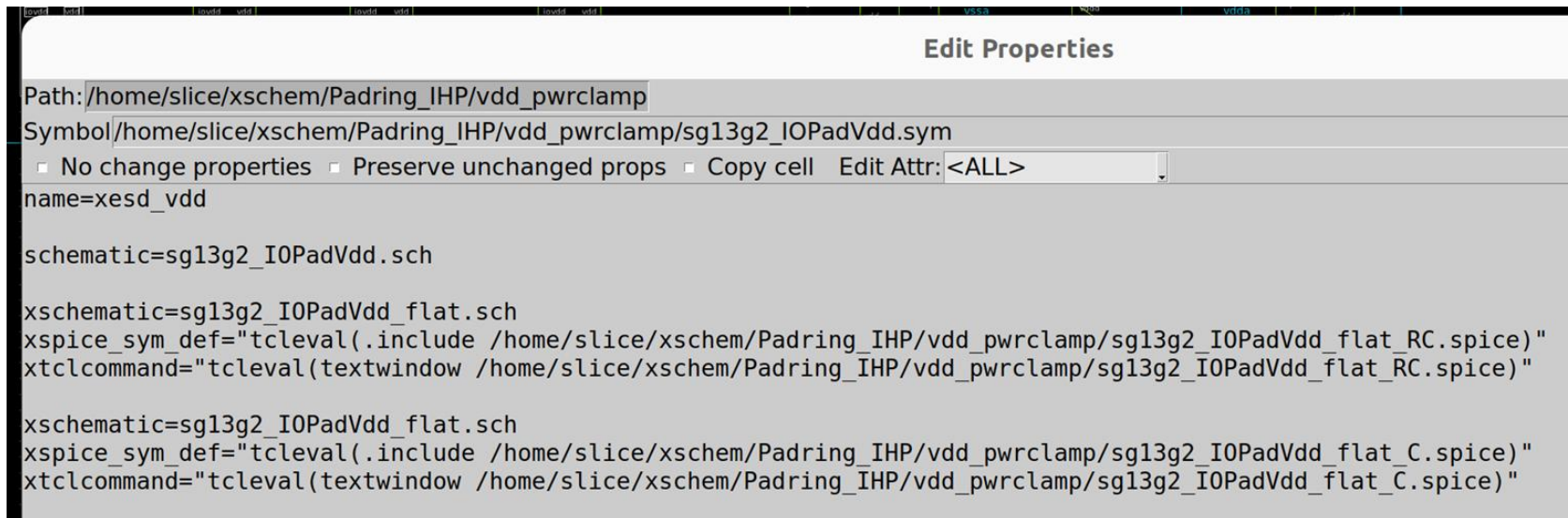
Vdd pwr clamp: PEX

- PEX C / RC files have been checked into the git repo as:

 sg13g2_IOPadVdd_flat_C.spice

 sg13g2_IOPadVdd_flat_RC.spice

- These can be cloned into a local area and run in xschem with instance binding similar to the below:



```
Path: /home/slice/xschem/Padring_IHP/vdd_pwrclamp
Symbol: /home/slice/xschem/Padring_IHP/vdd_pwrclamp/sg13g2_IOPadVdd.sym
  No change properties  Preserve unchanged props  Copy cell  Edit Attr: <ALL>
name=xesd_vdd

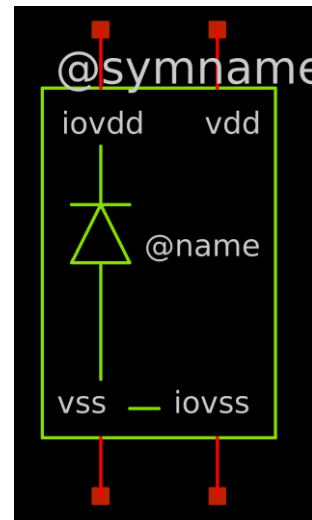
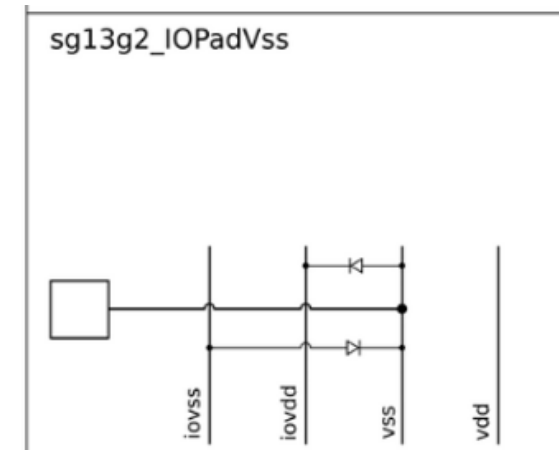
schematic=sg13g2_IOPadVdd.sch

xschematic=sg13g2_IOPadVdd_flat.sch
xspice_sym_def="tcleval(.include /home/slice/xschem/Padring_IHP/vdd_pwrclamp/sg13g2_IOPadVdd_flat_RC.spice)"
xtclcommand="tcleval(textwindow /home/slice/xschem/Padring_IHP/vdd_pwrclamp/sg13g2_IOPadVdd_flat_RC.spice)"

xschematic=sg13g2_IOPadVdd_flat.sch
xspice_sym_def="tcleval(.include /home/slice/xschem/Padring_IHP/vdd_pwrclamp/sg13g2_IOPadVdd_flat_C.spice)"
xtclcommand="tcleval(textwindow /home/slice/xschem/Padring_IHP/vdd_pwrclamp/sg13g2_IOPadVdd_flat_C.spice)"
```

Vss cell: Schematic

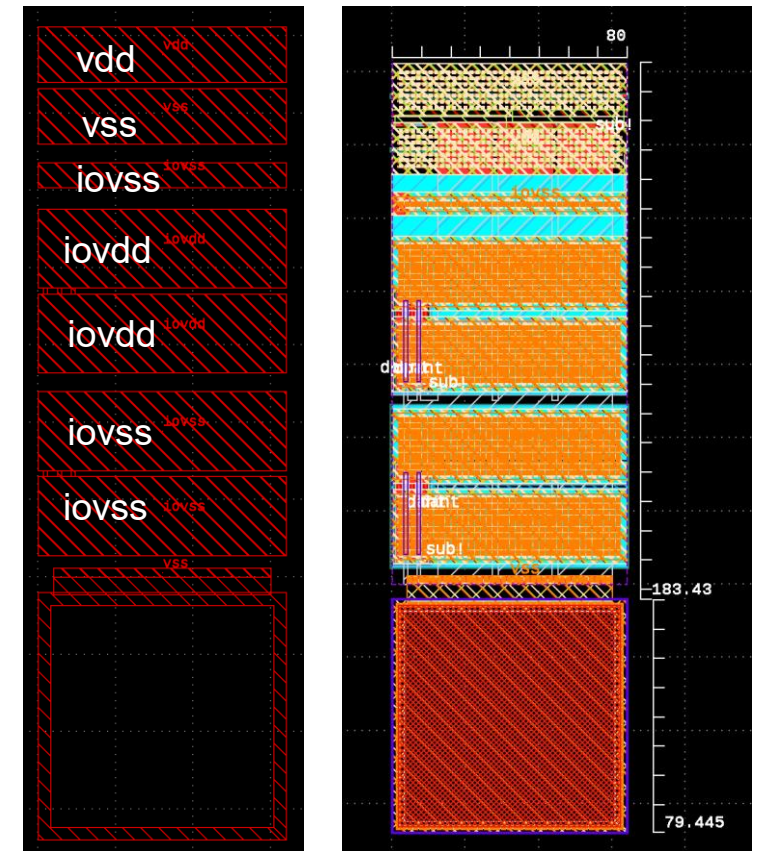
- Vss cell (with 80x80um bondpad) has been checked into:
- https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/vss_clamp
- Right shows the schematic drawing of the cell where it can be seen to be a basic cell consisting of a diode between iovss and iovdd
- There is also a diode between iovss and vss to reduce the delta between them and avoid cross domain logic errors
- Below shows the schematic symbol which can be found at the above link



Vss cell: Layout

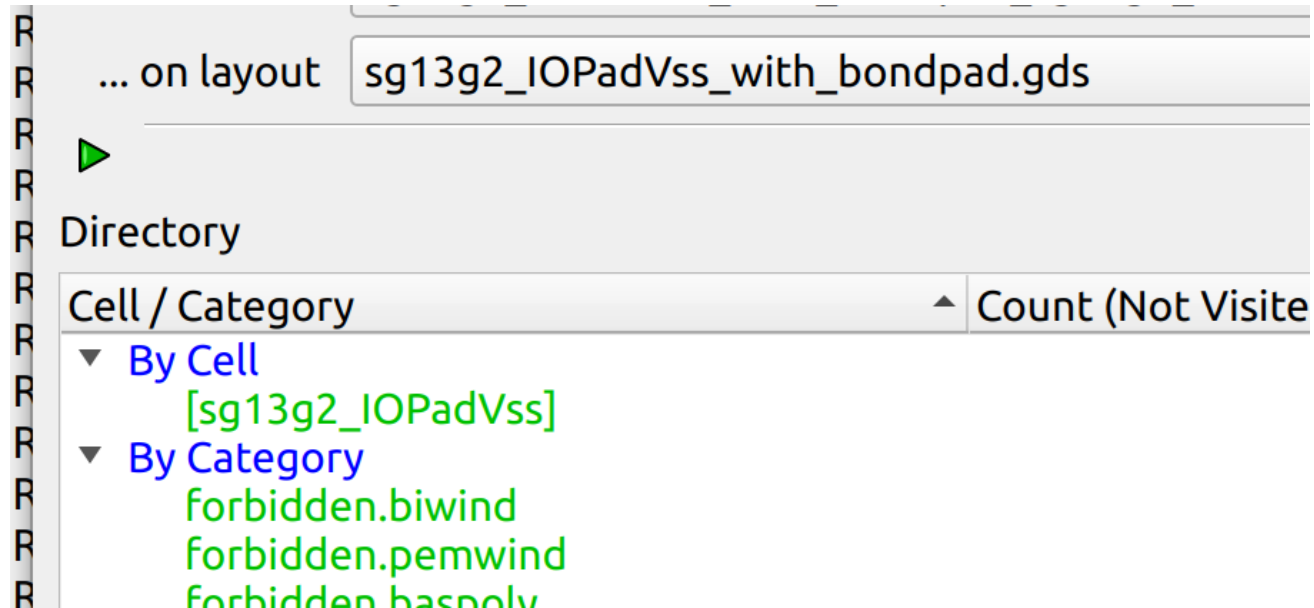
- Right shows the layout of the vddio pwr cell including the bondpad
- I/O cell footprint = 80 x 180um
- Bondpad footprint = 80 x 80um
- I/O cell + bondpad footprint = 80 x 260um
- Output pin occurs on M3
- M3 view clearly shows the supply connections
- gds can be found in the git repo as:

 sg13g2_IOPadVss_with_bondpad.gds



Vss cell: DRC

- Block is DRC clean



Vss cell: PEX

- PEX C / RC files have been checked into the git repo as:

 sg13g2_IOPadVss_flat_C.spice

 sg13g2_IOPadVss_flat_RC.spice

- These can be cloned into a local area and run in xschem with instance binding similar to the below:

```
Path: /home/slice/xschem/Padring_IHP/vss_clamp
Symbol /home/slice/xschem/Padring_IHP/vss_clamp/sg13g2_IOPadVss.sym
  ■ No change properties  ■ Preserve unchanged props  ■ Copy cell  Edit Attr: <ALL>
name=xesd_vss

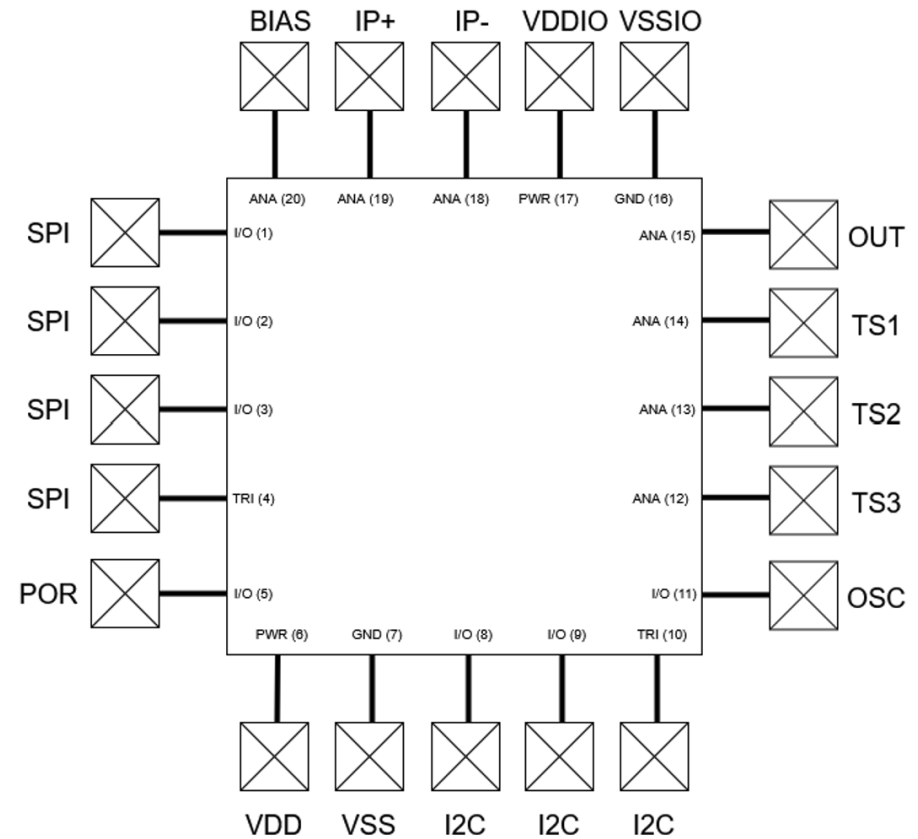
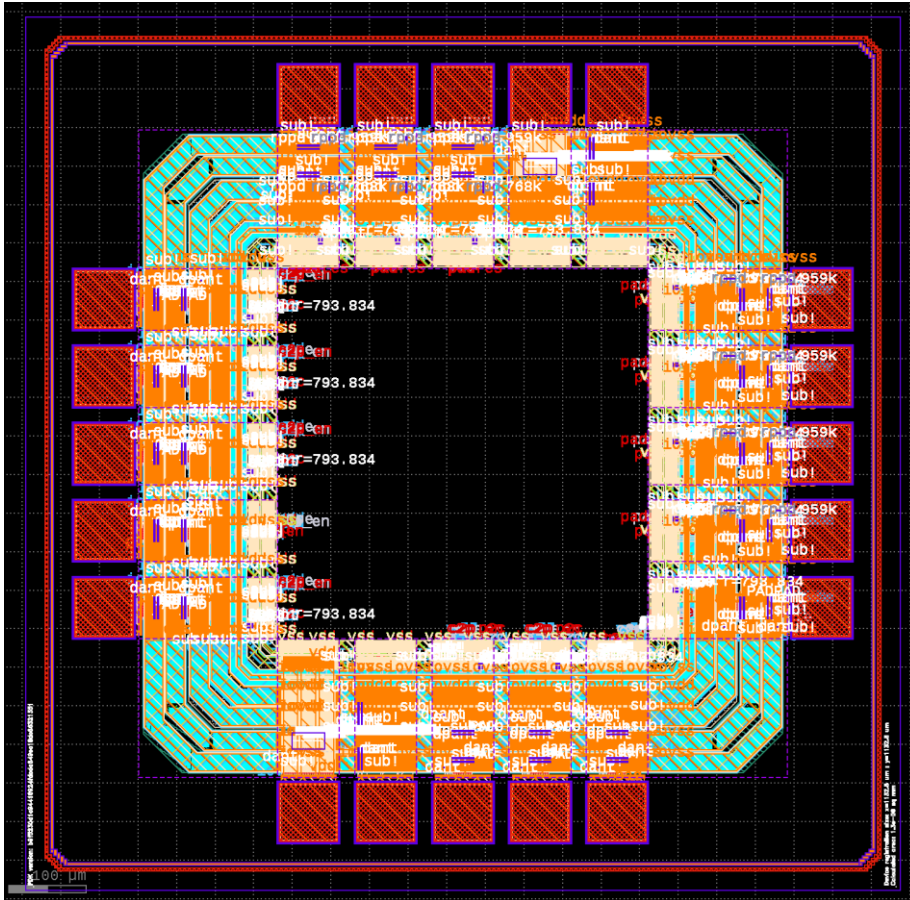
xschematic=sg13g2_IOPadVss.sch

xschematic=sg13g2_IOPadVss_flat.sch
xspice_sym_def="tclevel(.include /home/slice/xschem/Padring_IHP/vss_clamp/sg13g2_IOPadVss_flat_RC.spice)"
xtclcommand="tclevel(textwindow /home/slice/xschem/Padring_IHP/vss_clamp/sg13g2_IOPadVss_flat_RC.spice)"

schematic=sg13g2_IOPadVss_flat.sch
spice_sym_def="tclevel(.include /home/slice/xschem/Padring_IHP/vss_clamp/sg13g2_IOPadVss_flat_C.spice)"
tclcommand="tclevel(textwindow /home/slice/xschem/Padring_IHP/vss_clamp/sg13g2_IOPadVss_flat_C.spice)"
```

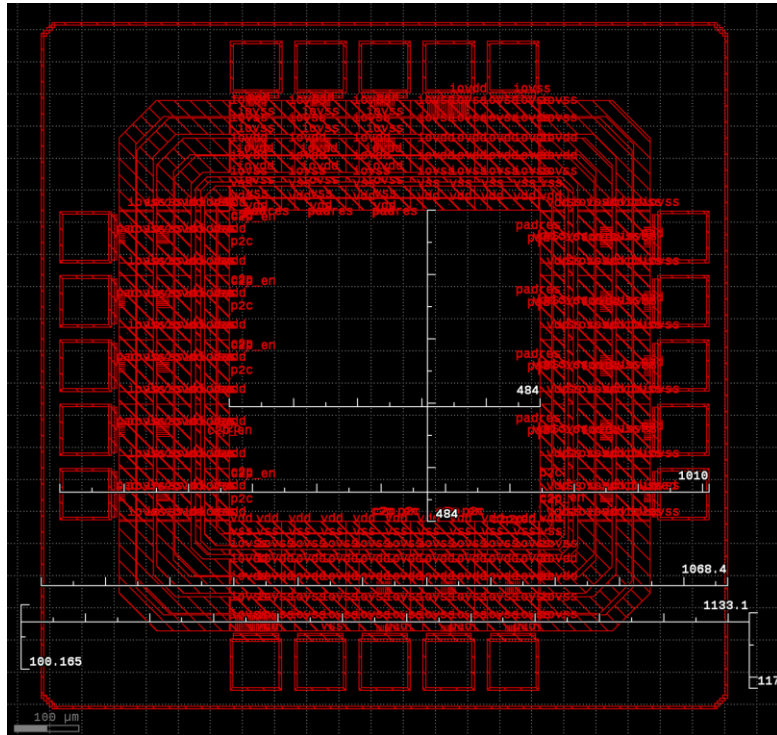

Complete padding

- Complete padding layout shown below. Padding illustration from slide 4 included for reference



Complete padding

- Padding dimensions including / excluding scribe line / seal ring, are shown below



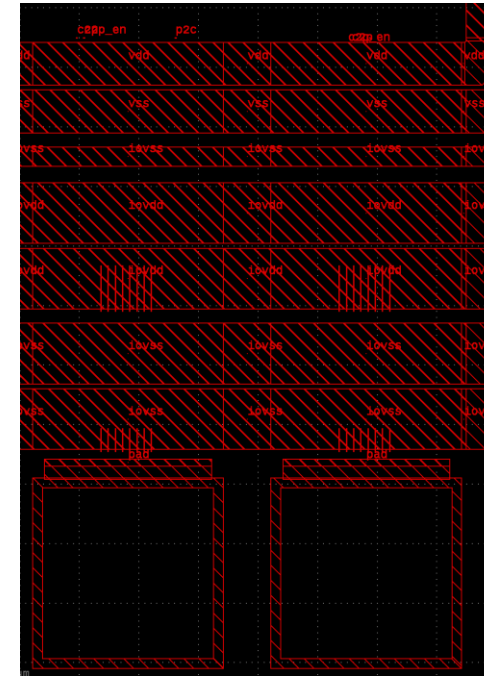
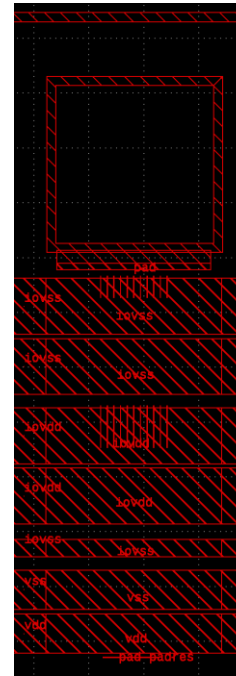
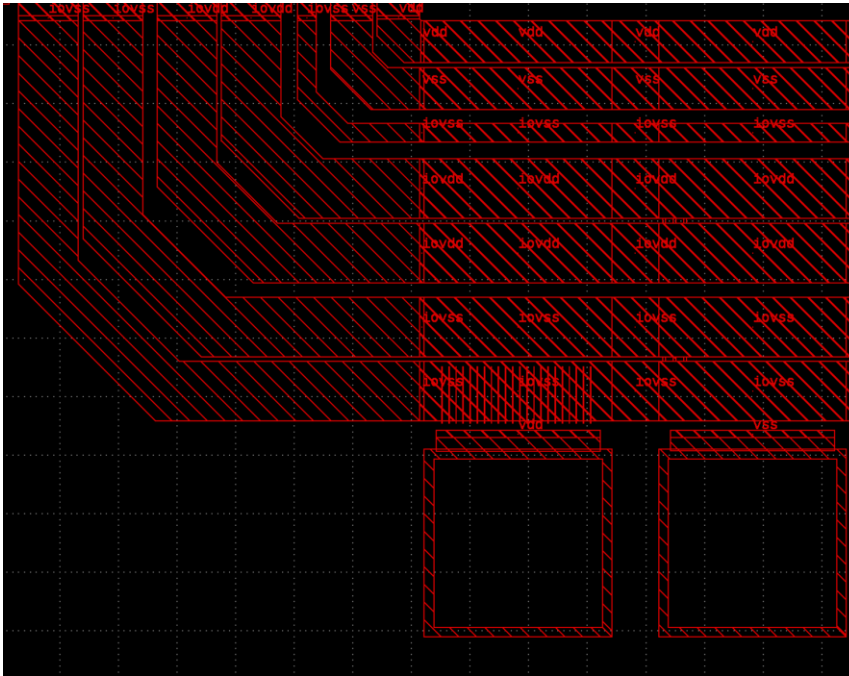
Reference	X (um)	Y (um)	A (mm2)
scribe line	1133	1133	1.28
seal ring	1068	1068	1.14
bond pad	1010	1010	1.02
design space	484	484	0.23

- Note: IHP defines area w.r.t seal ring. Therefore Sunrise ASIC area = 1.068mm2

- As shown, we are left with ~ 0.5 x 0.5mm area to fit the design into

Complete padding

- For clarity, below shows the power / gnd connections around the ring



- Full padding checked in at:
https://github.com/SLICESemiconductor/Sunrise/tree/main/analog/padring/padring_1mm2

Complete padding

- Only DRCs in the padding are the waivable DRCs from each analogIO cell
- This makes the padding effectively DRC clean

... on layout sunrise_pading.gds

▶

Directory

Cell / Category	Count (Not Visited) - Waived
▼ By Cell	28 (28)
[sunrise_pading]	
[sg13g2_IOPadAnalog]	
▶ [sg13g2_Clamp_P20N0D]	4 (4)
[sg13g2_IOPadAnalog\$1]	
▶ [sg13g2_Clamp_P20N0D\$1]	4 (4)
[sg13g2_IOPadAnalog\$2]	
▶ [sg13g2_Clamp_P20N0D\$2]	4 (4)
[sg13g2_IOPadAnalog\$3]	
▶ [sg13g2_Clamp_P20N0D\$3]	4 (4)
[sg13g2_IOPadAnalog\$4]	
▶ [sg13g2_Clamp_P20N0D\$4]	4 (4)
[sg13g2_IOPadAnalog\$5]	
▶ [sg13g2_Clamp_P20N0D\$5]	4 (4)
[sg13g2_IOPadAnalog\$6]	
▶ [sg13g2_Clamp_P20N0D\$6]	4 (4)
▼ By Category	28 (28)
forbidden hiwind	